

# MINGCHENG HE

University of Waterloo  
Electrical and Computer Engineering

Postdoctoral Fellow  
Email: mingcheng.he@uwaterloo.ca

## RESEARCH INTEREST

### **Satellite-terrestrial Integrated Networks**

Satellite communication system network modeling and performance evaluation  
Intelligent scheduling and protocol design for large-scale satellite networks  
Network slicing design and multi-dimensional resource management in heterogeneous networks  
Data-driven service demand prediction and traffic awareness

## ACADEMIC RESEARCH EXPERIENCE

### **Postdoctoral Fellow**

**February 2025 – Present**

Broadband Communications Research (BBCR) Laboratory, University of Waterloo, Canada

Supervisor: Professor Xuemin (Sherman) Shen and Professor Weihua Zhuang

- Research on satellite-terrestrial networks, artificial intelligence, and digital twin
  - Digital twin empowered resource scheduling for satellite-terrestrial networks
  - Artificial intelligence for network slicing

### **Research Assistant**

**September 2020 – December 2024**

Broadband Communications Research (BBCR) Laboratory, University of Waterloo, Canada

- Research on satellite-terrestrial integrated vehicular networks
  - Analyze network performance of satellite-terrestrial integrated vehicular network
  - Develop multi-dimensional resource scheduling schemes for the satellite-terrestrial integrated vehicular network
  - Investigate AI technologies for resource slicing in satellite-terrestrial integrated vehicular network

## EDUCATION

### **Doctor of Philosophy, Electrical and Computer Engineering**

**September 2020 – December 2024**

University of Waterloo, Waterloo, Ontario, Canada

Supervisor: Professor Xuemin (Sherman) Shen

Thesis: Slicing-Based Resource Management for Satellite-Terrestrial Integrated Vehicular Networks

### **Master of Engineering, Electronics and Communication Engineering**

**September 2017 – March 2020**

Shanghai Jiao Tong University, Shanghai, China

Supervisor: Professor Cunqing Hua

Thesis: Research on Concurrent Transmission Technology for Dual Connectivity Network

### **Exchange Program, Engineering Cycle**

**September 2017 – February 2018**

École Nationale Supérieure Des Mines de Paris (Mines Paris), Paris, France

### **Bachelor of Science, Information Engineering**

**September 2013 – August 2017**

Shanghai Jiao Tong University, Shanghai, China

## **SKILLS**

**C++/JAVA Programming**

**Data processing and machine learning (Python)**

**MATLAB**

**Network Simulator 3 (NS3)**

## **TEACHING EXPERIENCE**

**Teaching Assistant**

**September 2025 – December 2025**

University of Waterloo

Graduate course: Stochastic Processes (ECE604)

**Certificate of Expectations Teaching Assistant (TA) Training Workshop**

**September 2023**

University of Waterloo

The workshop involves:

- Four short lectures consisting of surprise and prepared topics
- A marking exercise under the guidance of three departmental mentors

**Teaching Assistant**

**February 2019 – June 2019**

Shanghai Jiao Tong University

Undergraduate course: Data Structure (CS137)

**Teaching Assistant**

**September 2018 – January 2019**

Shanghai Jiao Tong University

Undergraduate course: C Programming and Algorithm Analysis (CS139)

## **SELECTED SEMINARS AND PRESENTATIONS**

- [P1] Resource Slicing with Cross-Cell Coordination in Satellite-Terrestrial Integrated Networks  
**IEEE ICC 2024** — IEEE International Conference on Communications, Denver, CO, USA, June, 2024.
- [P2] Learning-Based Cache Placement and Content Delivery for Satellite-Terrestrial Integrated Networks  
**IEEE GLOBECOM 2021** — IEEE Global Communications Conference, Online, 2021.
- [P3] Design and Analysis for Dual Connectivity and Raptor Codes Assisted Handover in Vehicular Networks  
**IEEE WCNC 2020** — IEEE Wireless Communications and Networking Conference, Online, May, 2020.
- [P4] Global Traffic State Recovery VIA Local Observations with Generative Adversarial Networks  
**IEEE ICASSP 2020** — IEEE International Conference on Acoustics, Speech and Signal Processing, Online, May, 2020.

## **HONORS and AWARDS**

- Best Paper Award (first author), IEEE WCSP 2025.
- Best Paper Award (first author), IEEE GLOBECOM 2024.
- Second Place in Four Minute Thesis (4MT) Competition, IEEE ComSoc 2024.
- Faculty of Engineering (FoE) Award, University of Waterloo, 2021, 2024.
- University of Waterloo Graduate Scholarship, University of Waterloo, 2023 – 2024.
- Xuemin Shen Graduate Scholarship in Telecommunication, University of Waterloo, 2022, 2023.
- Jon W. Mark Graduate Scholarship in Communication, University of Waterloo, 2021.
- Ontario Research & Development Challenge Fund (ORDCF) Bell Scholarship, University of Waterloo, 2021 – 2024.
- International Doctoral Student Award, University of Waterloo, 2020 – 2023.
- Huawei Graduate Entrance Awards, University of Waterloo, 2020.
- Academic Excellence Scholarship (First-Class), Shanghai Jiao Tong University, 2019.

- Excellent Graduate of Shanghai Jiao Tong University, 2017.
- Academic Excellence Scholarship (Second-Class), Shanghai Jiao Tong University, 2016

## PUBLICATION

### Journal Papers

- [J1] **M. He**, H. Wu, C. Zhou, S. Hu, X. Shen, and W. Zhuang, "Adaptive Multi-Dimensional Resource Slicing in Cognitive Satellite-Terrestrial Vehicular Networks," **IEEE Transactions on Cognitive Communications and Networking**, vol. 12, pp. 3191-3208, 2026.
- [J2] **M. He**, H. Wu, X. Shen, and W. Zhuang, "Cooperative Resource Scheduling for Environment Sensing in Satellite-Terrestrial Vehicular Networks," **IEEE Internet of Things Journal**, vol. 12, no. 6, pp. 6734-6748, 2025.
- [J3] **M. He**, C. Hua, W. Xu, P. Gu, and X. Shen, "Delay Optimal Concurrent Transmissions with Raptor Codes in Dual Connectivity Networks," **IEEE Transactions on Network Science and Engineering**, vol. 8, no. 2, pp. 1478-1491, 2021.
- [J4] J. Li, **M. He**, C. Zhou, X. Huang, Z. Liu, L. Zhao, C.-X. Wang, and H. Wu, "Integration of Generative AI and Mobile Networking: A Comprehensive Survey," (invited paper), **IEEE Transactions on Network Science and Engineering**, vol. 13, pp. 4369-4405, 2026. (Corresponding Author)
- [J5] Y. Pei, **M. He**, H. Wu, X. Huang, C. Zhou, W. Zhuang, and X. Shen, "Mobility-Aware Resource Provisioning for Edge-Assisted Extended Reality Services," **IEEE Internet of Things Journal**, vol. 13, no. 2, pp. 2331-2344, 2026.
- [J6] S. Hu, J. Gao, X. Qin, C. Zhou, X. Huang, M. Li, **M. He**, and X. Shen, "Drift-Adaptive Slicing-Based Resource Management for Cooperative ISAC Networks," **IEEE Transactions on Cognitive Communications and Networking**, vol. 12, pp. 2631-2646, 2026.
- [J7] J. Yu, C. Hua, L. Liu, P. Gu, and **M. He**, "Accurate Beam Tracking for Robust USV-to-Satellite Transmission under Wave Fluctuation," **IEEE Transactions on Wireless Communications**, submitted, 2025.
- [J8] M. Ouyang, J. Liu, K. Qu, **M. He**, C. Zhou, B. Wang, R. Zhang, and W. Zhuang, "Intelligent Multipath Planning and Traffic Scheduling for Integrated Satellite-Terrestrial Networks," **IEEE Transactions on Network Science and Engineering**, major revision, 2025.
- [J9] Z. Gong, J. Yu, P. Gu, L. Liu, **M. He**, and C. Hua, "Virtual Network Embedding and Dynamic Traffic Scheduling for LEO Satellite Networks," **IEEE Transactions on Network and Service Management**, submitted, 2025.
- [J10] X. Huang, H. Yang, C. Zhou, **M. He**, X. Shen, and W. Zhuang, "When Digital Twin Meets Generative AI: Intelligent Closed-Loop Network Management," **IEEE Network**, Vol. 39, No. 5, pp. 272-279, 2025.
- [J11] H. Wu, **M. He**, X. Shen, W. Zhuang, N.-D. Dao, and W. Shi, "Network Performance Analysis of Satellite-Terrestrial Vehicular Network," **IEEE Internet of Things Journal**, vol. 11, no. 9, pp. 16829-16844, 2024.
- [J12] X. Shen, J. Gao, C. Zhou, S. Hu, **M. He**, and W. Zhuang, "Toward Immersive Communications in 6G," **Frontiers in Computer Science**, vol. 4, pp. 1-45, 2023.
- [J13] Z. Wang, H. Zhu, **M. He**, Y. Zhou, X. Luo, and N. Zhang, "GAN and Multi-Agent DRL Based Decentralized Traffic Light Signal Control," **IEEE Transactions on Vehicular Technology**, vol. 71, no. 2, pp. 1333-1348, 2021.

### Refereed Conference Papers

- [C1] **M. He**, Y. Pei, S. Hu, Z. Tang, W. Zhuang, and X. Shen, "Adaptive Resource Provisioning in Satellite Networks via VAE-Assisted Contextual Bandit Learning," **WCSP 2025** — International Conference on Wireless Communications and Signal Processing, Chongqing, China, 2025. (Best Paper Award)

- [C2] **M. He**, H. Wu, C. Zhou, S. Hu, Z. Tang, and W. Zhuang, “Digital Twin-Assisted Robust and Adaptive Resource Slicing in LEO Satellite Networks,”  
**GLOBECOM 2024** — IEEE Global Communications Conference, Cape Town, South Africa, 2024. (**Best Paper Award**)
- [C3] **M. He**, H. Wu, C. Zhou, and X. Shen, “Resource Slicing with Cross-Cell Coordination in Satellite-Terrestrial Integrated Networks,”  
**ICC 2024** — IEEE International Conference on Communications, Denver, CO, USA, 2024.
- [C4] **M. He**, C. Zhou, H. Wu, and X. Shen, “Learning-Based Cache Placement and Content Delivery for Satellite-Terrestrial Integrated Networks,”  
**GLOBECOM 2021** — IEEE Global Communications Conference, Madrid, Spain, 2021.
- [C5] **M. He**, C. Hua, and P. Gu, “Design and Analysis for Dual Connectivity and Raptor Codes Assisted Handover in Vehicular Networks,”  
**WCNC 2020** — IEEE Wireless Communications and Networking Conference, Seoul, South Korea, 2020.
- [C6] **M. He**, X. Luo, Z. Wang, F. Yang, H. Qian, and C. Hua, “Global Traffic State Recovery via Local Observations with Generative Adversarial Networks,”  
**ICASSP 2020** — IEEE International Conference on Acoustics, Speech and Signal Processing, Barcelona, Spain, 2020.
- [C7] Y. Pei, **M. He**, S. Hu, H. Hashida, W. Zhuang, and X. Shen, “Joint Rendering Quality and Encoding Type Selection for Edge-Assisted Extended Reality,”  
**ICC 2026** — IEEE International Conference on Communications, Glasgow, UK, 2026.
- [C8] Z. Gong, J. Yu, S. Hu, P. Gu, L. Liu, **M. He**, and C. Hua, “Virtual Network Embedding based Traffic Scheduling for LEO Satellite Constellations,”  
**GLOBECOM 2025** — IEEE Global Communications Conference, Taipei, Taiwan, 2025.
- [C9] X. Shao, Y. Zhang, S. Hu, Z. Tang, **M. He**, X. Huang, W. Zhuang, and X. Shen, “Hybrid-Field 6D Movable Antenna for Terahertz Communications: Channel Modeling and Estimation,”  
**ICCC 2025** — IEEE/CIC International Conference on Communications in China, Shanghai, China, 2025.
- [C10] Y. Pei, S. Hu, C. Zhou, X. Huang, **M. He**, W. Zhuang, and X. Shen, “Interaction-Aware Resource Reservation for Edge-Assisted Mobile Extended Reality,”  
**ICCC 2025** — IEEE/CIC International Conference on Communications in China, Shanghai, China, 2025.
- [C11] S. Hu, J. Gao, X. Huang, C. Zhou, **M. He**, and X. Shen, “Model Drift-Adaptive Resource Reservation in ISAC Networks: A Digital Twin-Based Approach”  
**ICCC 2024** — IEEE/CIC International Conference on Communications in China, Hangzhou, China, 2024.
- [C12] C. Zhou, H. Wu, **M. He**, W. Wu, N. Cheng, and X. Shen, “Adaptive Access Mode Selection in Space-Ground Integrated Vehicular Networks,”  
**GLOBECOM 2021** — IEEE Global Communications Conference, Madrid, Spain, 2021.
- [C13] X. Gong, Y. Zhou, Y. Bi, **M. He**, S. Sheng, H. Qiu, R. He, and J. Lu, “Estimating Web Attack Detection via Model Uncertainty from Inaccurate Annotation,”  
**CSCloud/EdgeCom 2019** — IEEE International Conference on Cyber Security and Cloud Computing/IEEE International Conference on Edge Computing and Scalable Cloud, Paris, France, 2019.

## **PROFESSIONAL SERVICES**

### **Technical Program Committee**

IEEE Global Communications Conference (GLOBECOM) 2026  
 International Conference on Wireless Communications and Signal Processing (WCSP) 2025  
 IEEE Global Communications Conference (GLOBECOM) 2025  
 IEEE International Conference on Computing, Networking and Communications (ICNC) 2025  
 IEEE Global Communications Conference (GLOBECOM) 2024  
 IEEE Vehicular Technology Conference Fall (VTC-Fall) 2024  
 IEEE Vehicular Technology Conference Fall (VTC-Fall) 2023  
 IEEE International Conference on Computer Communications Workshop (INFOCOM Workshop) 2022

**Reviewer of Journals**

IEEE Journal on Selected Areas in Communications (JSAC)  
IEEE Transactions on Wireless Communications (TWC)  
IEEE Transactions on Mobile Computing (TMC)  
IEEE Transactions on Cognitive Communications and Networking (TCCN)  
IEEE Transactions on Vehicular Technology (TVT)  
IEEE Internet of Things Journal (IoTJ)  
IEEE Open Journal of the Communications Society (OJCOMS)  
IEEE Communications Magazine  
IEEE Access  
IEEE Network  
Springer Peer-to-Peer Networking and Applications  
Wireless Networks  
China Communications  
Electronics and Telecommunications Research Institute (ETRI) Journal

**Reviewer of Conferences**

IEEE Global Communications Conference (GLOBECOM)  
IEEE International Conference on Communications (ICC)  
IEEE Vehicular Technology Conference (VTC)  
IEEE International Conference on Distributed Computing Systems (ICDCS)  
IEEE International Conference on Computer Communications Workshop (INFOCOM Workshop)  
IEEE Vehicular Technology Conference Workshop (VTC Workshop)  
IEEE/CIC International Conference on Communications in China (ICCC)  
IEEE Wireless Communications and Networking Conference (WCNC)  
IEEE Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)  
IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)  
IEEE International Performance Computing and Communications Conference (IPCCC)  
IEEE International Conference on Computing, Networking and Communications (ICNC)  
IEEE International Conference on High Performance Computing and Communications (HPCC)  
IEEE International Conference on Communication Technology (ICCT)  
IEEE/CIC International Conference on Communications in China Workshop (ICCC Workshop)